STAT201 · STATISTICS · LECTURE 3

Conditional Probability and Bayes' Theorem

How to update probabilities with new data, and why an “accurate” test can be wrong most of the time

Syllabus topic 3 · week 3 · 4 academic hours

LECTURE GOALS

- compute conditional probability from the definition and from a contingency table;
- apply the law of total probability and build a probability tree;
- derive and apply Bayes’ theorem, expressing the result through natural frequencies;
- explain why the base rate (the share of a rare event) determines the result, and not confuse $P(A \mid B)$ with $P(B \mid A)$;
- update a probability sequentially from several signals and understand when the signals may be treated as independent;
- make a decision by a threshold, comparing expected losses.

1 Conditional probability

In Lecture 2 we defined conditional probability through the intersection. Let us recall the definition and understand what it means.

CONDITIONAL PROBABILITY

The probability of event A given that event B has occurred:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0. \tag{1.1}$$

The meaning of the formula: having learned that B occurred, we *narrow* the sample space to B. The denominator becomes $P(B)$, and the numerator keeps only the outcomes lying in both A and B. The event after the bar determines the new denominator.

EXAMPLE 1. MOBILE APP AND DEPOSITS

Of 1 000 bank customers, 400 use the mobile app (M); 250 of them opened a deposit (D). Among the other 600, 120 opened a deposit.

	Deposit (D)	No deposit (D')	Total
App (M)	250	150	400
No app (M')	120	480	600
Total	**370**	**630**	**1 000**

$P(D) = 370/1000 = 0.37$. $P(D \mid M) = 250/400 = 0.625$. $P(D \mid M') = 120/600 = 0.20$.

App users open a deposit three times as often. And now the reverse probability: $P(M \mid D) = 250/370 = 0.676$ — this is the share of app users *among those who opened a deposit*. It is a different number with a different denominator.

$P(A \mid B) \neq P(B \mid A)$

In general $P(A \mid B) \neq P(B \mid A)$. The probability that a person with a fever has the flu and the probability that a person with the flu has a fever are different numbers. Mixing them up is the most common mistake in working with probabilities; this whole lecture is about how to move correctly from one to the other.

The definition immediately gives the multiplication rule $P(A \cap B) = P(A)\,P(B \mid A) = P(B)\,P(A \mid B)$ (Lecture 2). Two events are independent if and only if $P(A \mid B) = P(A)$.

2 The law of total probability

Often an event B can come about in several mutually exclusive ways. Events $A_1, \dots, A_k$ are said to form a *partition* of the sample space if they are pairwise mutually exclusive and together make up all of S ($A_1 \cup \dots \cup A_k = S$).

LAW OF TOTAL PROBABILITY

$$P(B) = \sum_{i=1}^{k} P(B \mid A_i)\,P(A_i) = P(B \mid A_1)P(A_1) + \dots + P(B \mid A_k)P(A_k). \qquad (2.1)$$

The probability of B is a weighted average of the conditional probabilities, and the weights are the probabilities of the paths.

EXAMPLE 2. THREE MACHINES

A part is produced on machine A_1 (50% of output, 2% defective), on A_2 (30%, 3% defective) or on A_3 (20%, 5% defective). The probability that a random part is defective (B):

$$P(B) = 0.50 \cdot 0.02 + 0.30 \cdot 0.03 + 0.20 \cdot 0.05 = 0.010 + 0.009 + 0.010 = \mathbf{0.029}.$$

The overall defect rate of 2.9% lies between 2% and 5%, as it should for a weighted average.

3 Bayes' theorem

The law of total probability goes "from causes to effects": knowing each machine's defect rate, it gives the overall defect rate. Bayes' theorem solves the inverse problem: *a defective part was found; what is the probability it was produced by machine A_3?*

BAYES' THEOREM

From the two expressions for $P(A_i \cap B)$ we obtain

$$P(A_i \mid B) = \frac{P(B \mid A_i)\,P(A_i)}{P(B)} = \frac{P(B \mid A_i)\,P(A_i)}{\sum_j P(B \mid A_j)\,P(A_j)}. \tag{3.1}$$

The **prior** probability $P(A_i)$ is before observing B; the **posterior** $P(A_i \mid B)$ is after. The factor $P(B \mid A_i)$ is called the *likelihood*.

Derivation in one line: $P(A_i \mid B)\,P(B) = P(A_i \cap B) = P(B \mid A_i)\,P(A_i)$; divide by $P(B)$ and substitute (2.1).

EXAMPLE 3. WHICH MACHINE IS TO BLAME?

The part is defective. The probability that it came from the third machine:

$$P(A_3 \mid B) = \frac{0.20 \cdot 0.05}{0.029} = \frac{0.010}{0.029} = \mathbf{0.345}.$$

Similarly $P(A_1 \mid B) = 0.010/0.029 = 0.345$ and $P(A_2 \mid B) = 0.009/0.029 = 0.310$. The sum is $0.345 + 0.310 + 0.345 = 1$. Although machine A_3 makes only 20% of the parts, because of its high defect rate it is responsible for 34.5% of all defects.

3.1 The main example: an alert from a fraud-monitoring system

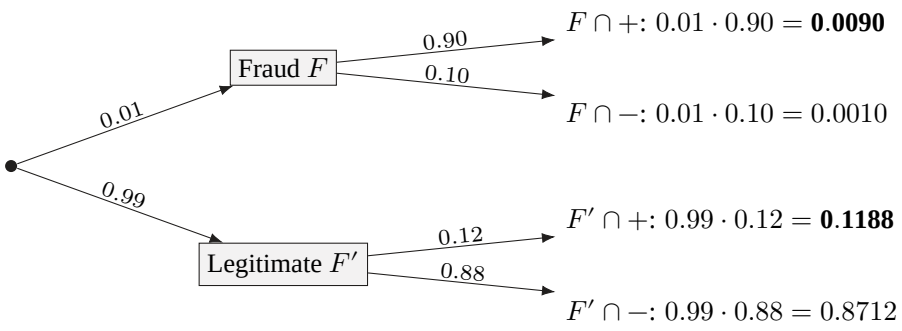
A bank uses a system that flags suspicious transactions. It is known that:

- **base rate**: 1% of transactions are fraudulent, $P(F) = 0.01$;
- **sensitivity**: the system catches 90% of fraudulent transactions, $P(+ \mid F) = 0.90$;
- **false-alarm rate**: the system wrongly flags 12% of legitimate transactions, $P(+ \mid F') = 0.12$.

A transaction is flagged. What is the probability that it is really fraudulent, $P(F \mid +)$?

Before computing, estimate by eye: is the answer closer to 7%, 50% or 90%? Most people say 90%.

The probability tree.



The alert occurs in two leaves: $P(+) = 0.0090 + 0.1188 = 0.1278$. Then

$$P(F \mid +) = \frac{0.0090}{0.0090 + 0.1188} = \frac{0.0090}{0.1278} = \mathbf{0.0704} \approx 7.0\%.$$

Natural frequencies (the same calculation without probabilities; it is usually more vivid). Take 10 000 transactions.

	Flagged	Not flagged	Total
Fraudulent (1%)	$100 \cdot 0.90 = 90$	10	100
Legitimate (99%)	$9\,900 \cdot 0.12 = 1\,188$	8 712	9 900
Total	**1 278**	**8 722**	**10 000**

Of the 1 278 flagged transactions only 90 are really fraudulent, i.e. $90/1\,278 = 7.04\%$. The system catches 90% of fraudsters, but almost 93% of its signals are false.

THE BASE RATE DECIDES EVERYTHING

Fraud is rare (1%), so there are a hundred times as many legitimate transactions. Even a small false-alarm rate (12%), applied to the huge mass of legitimate transactions, produces more signals than the 90% sensitivity applied to the small mass of fraudsters. Neglecting the base rate is called the *base rate fallacy*.

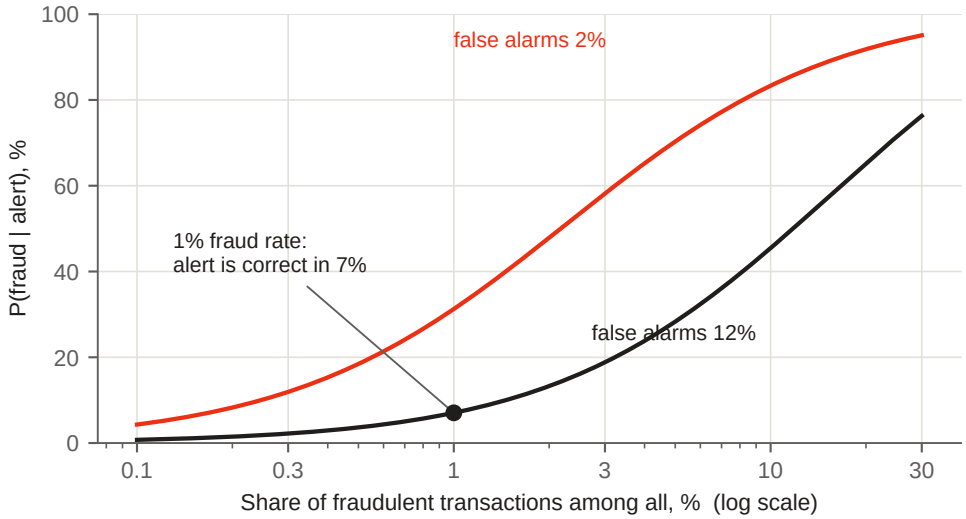


Fig. 1. Probability of fraud after an alert as a function of the base rate (sensitivity 90%). With 1% fraud and 12% false alarms the alert is right in only 7% of cases; reducing false alarms to 2% raises the answer to 31%.

Term	Notation	Meaning
Sensitivity	$P(+ \mid F)$	the share of real cases that the test finds
Specificity	$P(- \mid F') = 1 - P(+ \mid F')$	the share of legitimate transactions that the test lets through (here 0.88)
Positive predictive value	$P(F \mid +)$	the share of correct signals among all signals (here 0.0704)

3.2 An algorithm for Bayes problems

1. Name the events and write out *which* conditional probability is being asked for.
2. Write out the knowns: prior probabilities and likelihoods.
3. Build the tree (or a table of 10 000 objects) and find the leaf probabilities.
4. Add up the leaves in which the condition holds and divide the required leaf by this sum.
5. Check: the posterior probabilities over all hypotheses sum to 1; compare the answer with the prior probability.

4 Sequential updating and the odds form

Today's posterior probability becomes tomorrow's prior: new evidence arrives and we apply Bayes' theorem again. For such chains the notation through *odds* is convenient.

BAYES' RULE IN ODDS FORM

The odds of an event F: $\text{odds}(F) = \dfrac{P(F)}{1-P(F)}$; conversely, $P(F) = \dfrac{\text{odds}}{1+\text{odds}}$. Then

$$\underbrace{\frac{P(F \mid +)}{P(F' \mid +)}}_{\text{posterior odds}} = \underbrace{\frac{P(F)}{P(F')}}_{\text{prior odds}} \times \underbrace{\frac{P(+ \mid F)}{P(+ \mid F')}}_{LR_+}, \qquad LR_+ = \frac{\text{sensitivity}}{\text{false-alarm rate}}. \tag{4.1}$$

The likelihood ratio LR_+ shows how many times more likely a positive result is with fraud than without it.

EXAMPLE 4. ONE AND TWO ALERTS

Here $LR_+ = 0.90/0.12 = 7.5$. The prior odds are $0.01/0.99 = 0.0101$.
After one alert: $0.0101 \cdot 7.5 = 0.0758$, hence $P = 0.0758/1.0758 = 0.0704$. The answer coincides with the calculation above.
After a second independent alert: $0.0101 \cdot 7.5^2 = 0.568$, hence $P = 0.568/1.568 = \mathbf{0.362}$.
Conclusion: two independent signals raise the probability from 7% to 36%, but not to 90%.

DOUBLE COUNTING

Likelihood ratios may be multiplied only if the signals are *conditionally independent* given the state (under fraud and without it). Two models trained on the same customer data often output the same thing, and the second alert carries almost no new information. An automatic model and a subsequent document check are much closer to independent. In the extreme case, if the second alert simply repeats the first, the probability remains 7.04%, not 36.2%.

5 A decision by threshold

A probability by itself makes no decision. The costs of errors must be compared. Let the possible loss from fraud be $L = 10$ million UZS and the cost of an investigation $c = 0.5$ million UZS. Investigating

pays off if the *expected loss prevented* exceeds the cost of the investigation:

$$p \cdot L > c \quad \Longleftrightarrow \quad p > \frac{c}{L} = \frac{0.5}{10} = \mathbf{5}\%. \tag{5.1}$$

Both probabilities, 7.04% after one alert and 36.2% after two, exceed the 5% threshold, so an investigation is justified already after the first signal. Notice that the 5% threshold is far lower than the "intuitive" 50%: if the cost of a miss is high and checking is cheap, it is worth investigating even at a low probability.

6 A bridge to the next lecture: from events to random variables

An event says *what happened*. A random variable says *how much*. The same outcome can be evaluated by different numbers. For a single loan the outcomes "repaid", "overdue", "default" give the bank's profit X (million UZS):

Outcome	x, million UZS	$P(X = x)$
Repaid	$+1.2$	0.80
Overdue	$+0.4$	0.15
Default	-8.0	0.05

The average we expect in the long run is $\mathrm{E}[X] = 1.2 \cdot 0.80 + 0.4 \cdot 0.15 - 8 \cdot 0.05 = \mathbf{0.62}$ million UZS. This is *not* the profit on a particular loan, but the average over a set of similar loans. This apparatus is the subject of Lecture 4.

7 Key terms

Term	*In Russian*	Meaning
Conditional probability	*Условная вероятность*	$P(A \mid B) = P(A \cap B)/P(B)$
Partition	*Разбиение*	pairwise mutually exclusive events making up all of S
Law of total probability	*Формула полной вероятности*	$P(B) = \sum P(B \mid A_i)P(A_i)$
Bayes' theorem	*Теорема Байеса*	$P(A_i \mid B) = \dfrac{P(B \mid A_i)P(A_i)}{P(B)}$
Prior	*Априорная вероятность*	before the evidence is observed
Posterior	*Апостериорная вероятность*	after the evidence is observed
Likelihood	*Правдоподобие*	$P(\text{data} \mid \text{hypothesis})$
Base rate	*Базовая частота*	the share of the event in the original population
Sensitivity	*Чувствительность*	$P(+ \mid \text{present})$
Specificity	*Специфичность*	$P(- \mid \text{absent})$
Odds	*Шансы*	$p/(1-p)$
Likelihood ratio	*Отношение правдоподобия*	$LR_+ = \text{sens.}/\text{false-alarm rate}$

Formulas of Lecture 3

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) = \sum_i P(B \mid A_i)P(A_i),$$

$$P(A_i \mid B) = \frac{P(B \mid A_i)P(A_i)}{\sum_j P(B \mid A_j)P(A_j)},$$

$$\text{odds:}\ \ \text{odds} = \frac{p}{1-p}, \quad p = \frac{\text{odds}}{1+\text{odds}}, \quad \text{odds}_{\text{post}} = \text{odds}_{\text{prior}} \times LR, \quad LR_+ = \frac{\text{sens.}}{\text{false-alarm rate}},$$

$$\text{investigate if } \ p > \frac{c}{L}.$$

8 Practice problems

Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix "Statistical Tables".

Problem 1 [A] It is known that $P(A \cap B) = 0.10$, $P(A) = 0.40$, $P(B) = 0.50$.

(a) Find $P(A \mid B)$ and $P(B \mid A)$.
(b) A student claims: "Since $P(A \mid B) = 0.8$, also $P(B \mid A) = 0.8$." Is he right?

Problem 2 [A] Of 1000 bank customers, 400 use the mobile app (M). Of these, 250 hold a deposit (D); among the other 600 customers, 120 hold a deposit.

(a) Find $P(\mathrm{D})$.
(b) Find $P(\mathrm{D} \mid \mathrm{M})$ and $P(\mathrm{D} \mid \overline{\mathrm{M}})$.
(c) Find $P(\mathrm{M} \mid \mathrm{D})$.

Problem 3 [B] A factory makes parts on three machines A, B, C with output shares of 50%, 30%, 20%. The defect rates are 2%, 3%, 5% respectively.

(a) Find the probability that a random part is defective.
(b) A part turns out to be defective. Find the probability that it was made on machine C; on A; on B.
(c) Check that the three posterior probabilities sum to 1.

Problem 4 [B] A bank catches fraudulent transactions. The share of fraudulent transactions is 2%. The system flags 85% of fraudulent transactions and wrongly flags 10% of normal ones.

(a) Find the probability that a flagged transaction is really fraudulent.
(b) Solve the same problem by the natural-frequency method on 10 000 transactions.
(c) Explain why the result is much smaller than the 85% sensitivity.

Problem 5 [B] A disease occurs in 0.5% of the population. A test detects it in 98% of the sick and gives a false positive in 5% of the healthy.

(a) What is the probability of disease given a positive result?
(b) The test was repeated independently (given health status) and is again positive. What is the probability of disease now?

Problem 6 [B] Continue the fraud problem (share 2%, sensitivity 85%, false alarms 10%). Suppose k independent (given fraud / no fraud) systems have raised a signal on the same transaction.

(a) Find the prior odds and the likelihood ratio LR^{+}.
(b) Find the probability of fraud for $k = 1,\ 2,\ 3$ signals.
(c) What happens to the result if the signals are not independent but fully duplicate each other?

Problem 7 [B] For a transaction in the fraud problem, the loss from a missed fraud is 8 million UZS, an investigation costs 0.4 million UZS (and completely eliminates the loss).

(a) Find the threshold probability above which an investigation pays off.
(b) Is it worthwhile to investigate a transaction that received one signal (fraud probability 14.8%)?
(c) At what minimum prior probability is one signal enough to justify an investigation?

Problem 8 [B] For a transaction with posterior fraud probability $p = 0.148$, three actions are considered: (1) let it pass (expected loss: $8p$ million); (2) investigate (cost 0.4 million, no loss); (3) block (if the transaction is normal, the damage to the customer is 1 million; if fraudulent, no loss). Which action minimizes the expected loss?

Problem 9 [C]

(a) Derive Bayes' formula for event A_i from the definition of conditional probability and the law of total probability (the events $A_1, \dots, A_n$ form a partition).

(b) Prove the odds form: $\dfrac{P(A \mid B)}{P(\overline{A} \mid B)} = \dfrac{P(A)}{P(\overline{A})} \cdot \dfrac{P(B \mid A)}{P(B \mid \overline{A})}$.

Problem 10 [C] A family has two children.

(a) It is known that at least one child is a boy. What is the probability that both are boys?

(b) It is known that the older child is a boy. What is the probability that both are boys?

Problem 11 [C] The Monty Hall problem. Behind one of three doors is a car, behind the other two are goats. You have chosen door 1. The host, who knows where the car is, opens a door with a goat (from the two remaining; if both have goats, he chooses at random) and offers you the chance to switch. Find the probability of winning if you switch.

Problem 12 [C] Two fraud-alert systems (sensitivity 90%, false alarms 12%, fraud share 1%) raised signals. An analyst multiplied the likelihood ratios and obtained 36.2%. It then turned out that the second system simply copies the decision of the first. What should the estimate of the probability be?